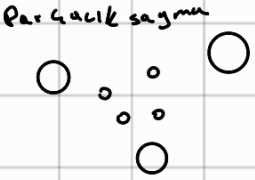
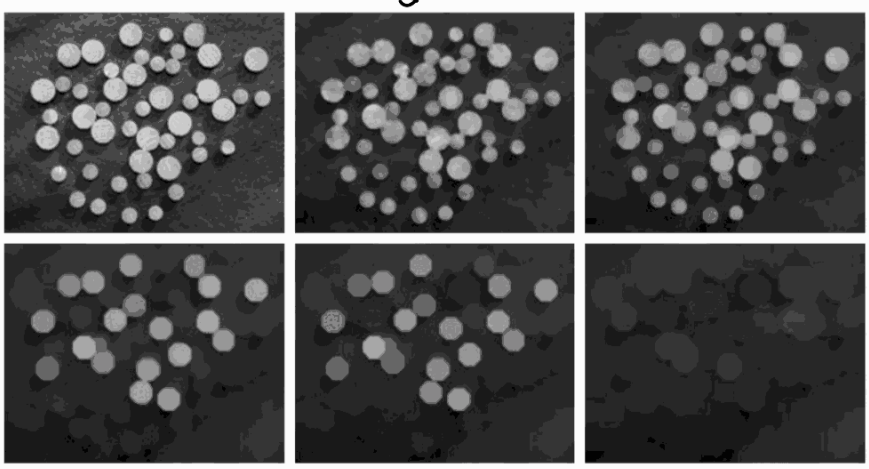
Soru cevap

Parçacık sayma

Granulometry



opening erosion, dialation



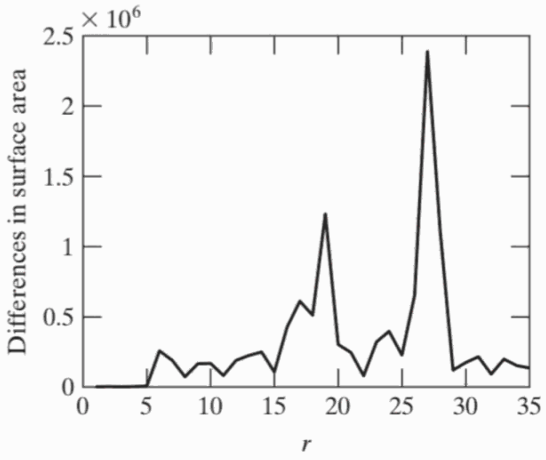
a b c
d e f

FIGURE 9.41 (a) 531 × 675 image of wood dowels. (b) Smoothed image. (c)–(f) Openings of (b) with disks of radii equal to 10, 20, 25, and 30 pixels, respectively. (Original image courtesy of Dr. Steve Eddins, The MathWorks, Inc.)

yapı elemanı her adımda büyütülür

Her pikselin değeri toplanır

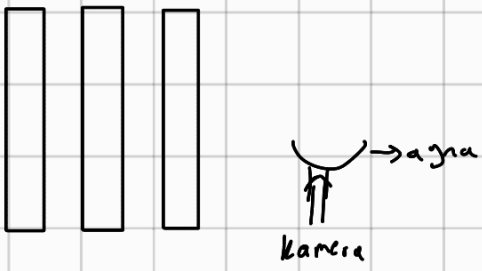
birbirinden çıkarırsak



ne büyüklükte objeler var

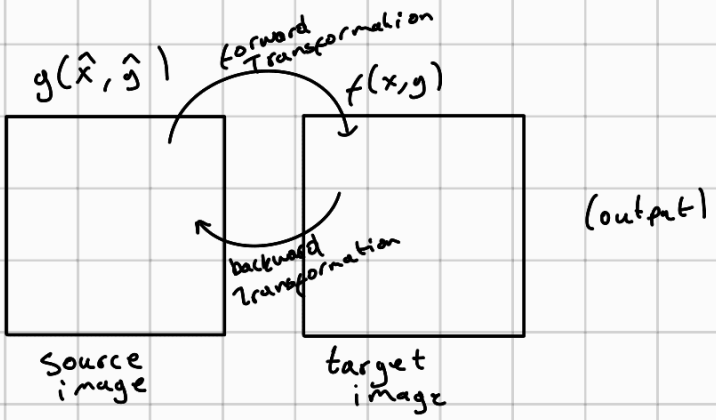
Router chef segmentasyon

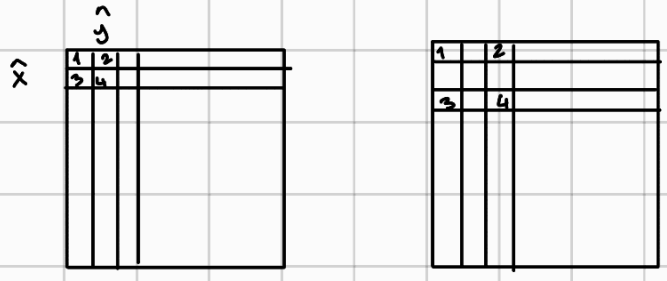
Geometric Transformation



hepsini aynı boyutta görmez

Geometric Transformation





psuedo-code

```
for x = 0; nrows in target
    for y = 0; ncols in target
        compute x̂ = inverse spatial
                    transformation of x
                ŷ = inverse spatial
                    Transformation y
        f(x,y) = gray level interpolation of g(x̂, ŷ)
```

c büyünce görüntü büyür

scaling: $\hat{x} = \frac{x}{c}$ $\hat{y} = \frac{y}{c}$

rotation: $\hat{x} = x\cos\theta - y\sin\theta$

$\hat{y} = x\sin\theta + y\cos\theta$

$$\begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Gray level interpolasyon

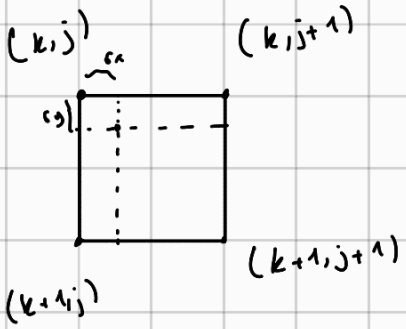
level 0 : nearest neighbour (NN)

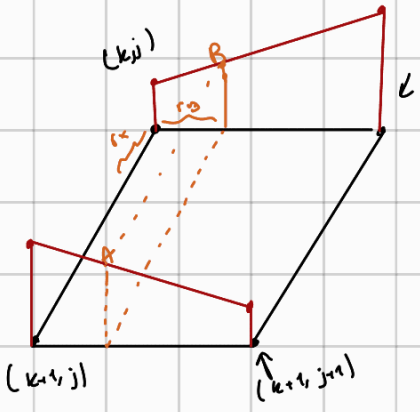
level 1 : bilinear interpolation (BI)

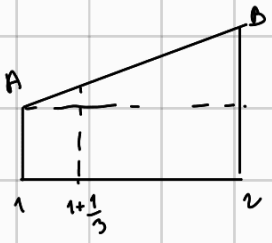
$\hat{x} = k + r_x$ $\hat{y} = j + r_y$

$\hat{x} = \frac{4}{3} = 1 + \frac{1}{3}$ ($k = 1$, $r_x = \frac{1}{3}$)

$\hat{y} = \frac{15}{2} = 2 + \frac{3}{6}$ ($k = 2$, $r_y = \frac{3}{6}$)







Örnek

	0	1	2	3
0	1	1	100	1
1	1	1	100	1
2	100	100	100	100
3	1	1	100	1

x \ y	0	1	2
0			
1			
2			

Sonuç

x \ y	0	1	2
0	1	34	34
1	34	56	56
2	34	56	56

$c = \frac{3}{4}$ $c = \frac{2}{3}$ alma

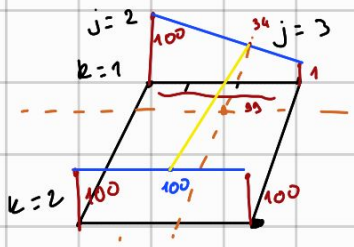
x	y	$\hat{x}$	$\hat{y}$	N N
0	0	0	0	0,0 1
0	1	0	4/3	0,1 1
0	2	0	8/3	0,3 1
1	0	4/3	0	1
1	1	4/3	4/3	1
1	2	4/3	8/3	1
2	0	8/3	0	1
2	1	8/3	4/3	1
2	2	8/3	8/3	1

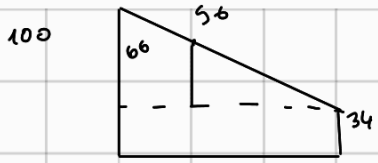
BI

$x = 1$ $y = 2$

$\hat{x} = \frac{4}{3}$ $\hat{y} = \frac{8}{3}$

$\hat{x} = 1 + \frac{1}{3}$ $\hat{y} = 2 + \frac{2}{3}$





Feature Extraction



Moment hesabı yapabilirsiniz

fourier discripter

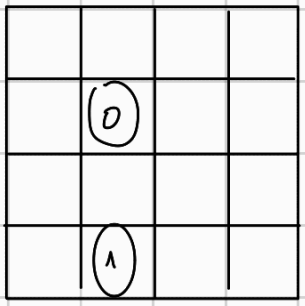


$$s(r) = x + jy$$

$$S[k] = \sum_{r=0}^{R} s(r)\, e^{-j 2\pi r k / R}$$

Texture analizi Co-occurance Matrix

$P_{\phi,d}(a,b)$



→ belli bir paterni var mı

